

# Direct Multi-Class Classification for Cuff-Less Hypertension Detection Using ECG and PPG Signals

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## Abstract

Cuff-less blood pressure (BP) monitoring has emerged as a promising alternative to conventional cuff-based measurement for continuous cardiovascular monitoring using wearable sensors. Most prior approaches estimate systolic and diastolic BP as continuous values and subsequently apply clinical thresholds, despite the fact that many real world decisions depend primarily on categorical risk stratification. In this study, we investigate whether direct multi-class classification of BP status provides a more effective framework for wearable hypertension screening. Using the UCI Cuff-Less Blood Pressure Estimation dataset containing synchronized electrocardiogram and photoplethysmogram recordings, we compare five machine learning approaches: Random Forest, XGBoost, 1D-CNN, a support vector regression (SVR) baseline with post-hoc thresholding, and a pulse transit time logistic regression reference model. All models were evaluated using subject-level cross validation and held-out testing to prevent information leakage across records. The best-performing model, the 1D-CNN, achieved an accuracy of 73.4%, macro F1 score of 0.686, and AUROC of 0.905 for hypertensive case detection, outperforming the conventional regression-based baseline by 0.223 AUROC and 31.3% in accuracy. These findings suggest that direct classification may offer a more clinically aligned and practically deployable approach for cuff-less hypertension detection from wearable physiological signals.

**Key Words:** Cuff-less blood pressure estimation; Hypertension classification; Electrocardiography (ECG); Photoplethysmography (PPG); Machine learning

## 1. Introduction

Hypertension is among the most prevalent and consequential cardiovascular risk factors worldwide, linked to critical conditions such as myocardial infarction, stroke, heart failure, chronic kidney disease, and premature mortality.<sup>1</sup> Early identification and consistent monitoring of elevated blood pressure (BP) are therefore central to preventive cardiovascular care. Conventional cuff-based BP measurements are intermittent, cumbersome for repeated daily use, and unsuitable for continuous wearable monitoring. As wearable health technologies continue to expand, there is increasing interest in cuff-less methods that estimate BP from non-invasive physiological signals such as electrocardiography (ECG) and photoplethysmography (PPG, used in pulse oximeters in clinical settings).<sup>2</sup>

Prior research in cuff-less BP monitoring has commonly approached the problem as a regression task, with models that predict continuous systolic and diastolic BP values from waveform-derived features such as pulse transit time. A notable example is the work of Kachuee et al., who demonstrated calibration-free BP estimation using pulse transit time and machine learning on the UCI Cuff-Less Blood Pressure Estimation dataset.<sup>2,3</sup> Accurate numerical estimation is valuable, but many real-world clinical decisions are based on threshold-defined categories such as normal, elevated, hypertensive, or hypotensive BP.<sup>1</sup> We investigate whether cuff-less BP assessment may be better framed as a classification problem using the ECG and PPG recordings from the UCI BP estimation dataset. We compare multiple machine learning approaches for categorizing

samples into clinically meaningful BP classes. By focusing on categorical risk identification rather than precise numerical derivation of BP, we aim to evaluate a more practical framework for wearable hypertension screening and cardiovascular risk monitoring.

## 2. Methods

### 2.1 Dataset and Study Design

In this study, we use the UCI Cuff-Less Blood Pressure Estimation dataset, which contains 11,987 synchronized ECG, PPG, and arterial blood pressure (ABP) records derived from the MIMIC-II waveform database.<sup>3</sup> This dataset is publicly available. Signals were sampled at 125 Hz and processed as fixed-length waveform segments. Reference systolic blood pressure (SBP) and diastolic blood pressure (DBP) values were obtained from the ABP waveform and used to assign categorical BP levels. The prediction task was formulated as a three-class classification problem based on 2017 ACC/AHA BP thresholds: Normal ( $<120/80$ ), Elevated ( $120\text{--}129/80$ ), and Hypertensive ( $\geq 130/80$ ).<sup>9</sup> The class distribution across the 128,144 segments was 47,362 Normal / 19,980 Elevated / 60,802 Hypertensive, in which the observed distribution is consistent with the data being obtained from ICU patients.

The UCI release does not preserve original MIMIC-II patient identifiers. Subject-level partitions in this study were therefore enforced at the record level, with each (part, record) tuple in the UCI release treated as a unique subject. This prevents within-record leakage of segments across train, validation, and test sets, but in the rare case that a single MIMIC-II patient contributed multiple records to the UCI release, patient-level disjointedness cannot be guaranteed; we discuss this further as a limitation in §4.5.

### 2.2 Preprocessing and Feature Construction

Raw ECG, PPG, and ABP waveforms were processed channel-by-channel prior to model development. PPG signals were filtered using a fourth-order Butterworth bandpass filter from 0.5 to 8 Hz with zero-phase application, while ECG signals were filtered using a fourth-order Butterworth bandpass filter from 0.5 to 40 Hz with zero-phase application.<sup>10</sup> ABP signals were left unfiltered to preserve systolic and diastolic peak amplitudes required for reference label derivation. Following filtering, each record was partitioned into non-overlapping 8-second windows (1,000 samples per segment). Each segment then underwent deterministic quality control, excluding segments containing NaN or infinite values, ABP values outside the physiologic range of 30 to 250 mmHg, ECG flat-line intervals longer than one second, or insufficient PPG variance. The final preprocessing pipeline yielded 128,144 valid waveform segments for downstream analysis.

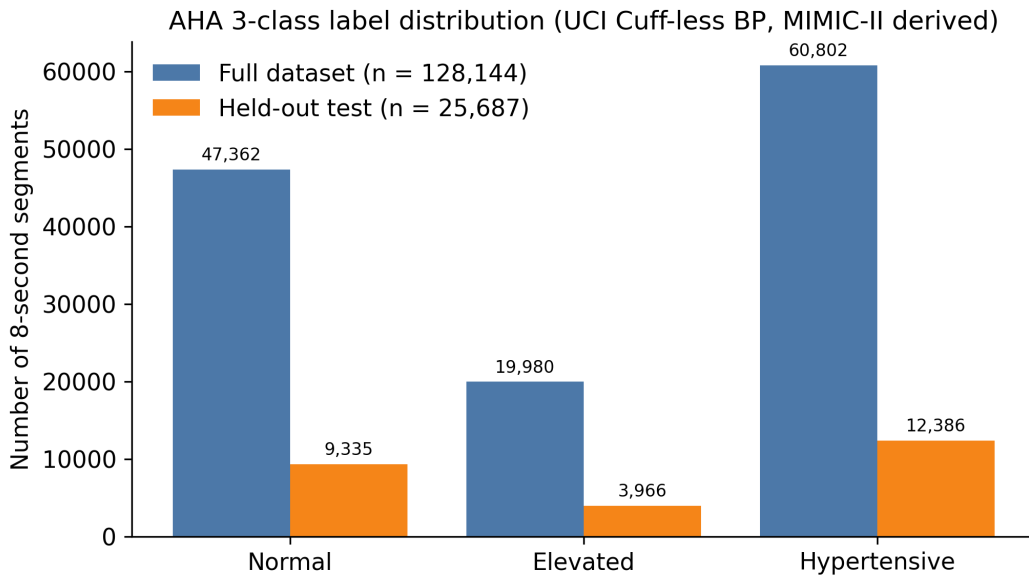
For tabular machine learning models, engineered physiological features were extracted from each valid segment. Two feature sets were used. The first reproduced the ten-feature set described by Kachuee et al. and was used for the support vector regression baseline. These variables included pulse transit time measures, heart rate, augmentation index, large-artery stiffness index, and four pulse waveform area descriptors. The second feature set, used for the Random Forest and XGBoost classifiers, expanded this representation to twenty features by adding heart rate variability metrics, pulse width measures, rise and decay timing, ECG QRS duration, amplitude ratios, and an additional augmentation-index variant. These added features were selected to capture broader cardiovascular dynamics and waveform morphology beyond

pulse transit time alone.

Some landmark-dependent features required detection of the diastolic peak or diastolic notch in the PPG waveform, which was inconsistently visible at the dataset sampling frequency. Rather than exclude incomplete segments and substantially reduce sample size, missing values were handled using median imputation within the model training pipeline, ensuring that imputation statistics were learned from training data only. For the convolutional neural network (CNN), no manual feature extraction was performed. Instead, synchronized ECG and PPG waveform segments were retained as paired two-channel inputs with shape (2, 1000), allowing end-to-end feature learning directly from the raw time series.

### 2.3 Data Splitting and Validation Strategy

To prevent record-level information leakage, all data partitions were performed at the record level rather than the segment level, with the limitation noted in §2.1. Subject identifiers were derived deterministically from the (part, record) indices provided in the UCI release, and each identifier was assigned to only one partition. As a result, no waveform segments from the same record were shared across training, validation, or testing. This represents an important methodological distinction from prior random segment-level splits, which can overestimate generalization performance when repeated segments from the same record appear in multiple sets. A common split definition was reused across all five models to ensure direct comparability of results.



**Figure 1.** Distribution of ACC/AHA three-class blood pressure labels in the full processed dataset and held-out test set. Similar class proportions were preserved after subject-level splitting, with hypertensive segments representing the largest class and elevated segments the smallest.

The final cohort was divided into training, validation, and held-out test sets consisting of 7,671 subjects (81,965 segments), 1,918 subjects (20,492 segments), and 2,398 subjects (25,687 segments), respectively. Model selection and hyperparameter tuning were performed using 5-fold GroupKFold cross-validation on the combined training and validation subjects, with subject identity used as the grouping variable in every fold. Final model performance was then evaluated once on the independent held-out test set containing subjects not observed during training or model selection. An internal leakage check was used to verify that no subject appeared in more

than one partition.

## **2.4 Model Development**

All five models were trained and evaluated using a shared subject-level split definition, ensuring that each method was assessed on the same held-out test cohort and under identical data partitioning conditions.

### **2.4.1 PTT-Only Logistic Regression**

A multinomial logistic regression model using pulse transit time as the sole predictor was implemented as a low-complexity reference baseline. The feature was standardized prior to training. This model was included to isolate the cross-subject predictive value of pulse transit time alone for hypertension classification.

### **2.4.2 Random Forest**

A Random Forest classifier was trained using the 20-feature engineered feature set.<sup>4</sup> Records with unresolved missing values were excluded for this model. Hyperparameters were optimized using 5-fold subject-level GroupKFold cross-validation across combinations of tree count, maximum depth, minimum leaf size, and feature subsampling strategy. The final selected model used 500 trees, a maximum depth of 20, minimum leaf size of 5, and square-root feature sampling at each split.

### **2.4.3 XGBoost**

An XGBoost multiclass classifier was trained on the same 20 engineered features.<sup>5</sup> Hyperparameter optimization was performed using 5-fold subject-level GroupKFold cross-validation over combinations of tree count, tree depth, learning rate, row subsampling, and column subsampling. The best-performing configuration used 500 estimators, maximum depth of 8, learning rate of 0.1, row subsampling of 0.8, and full column sampling. XGBoost was included because gradient boosting methods often achieve strong performance on structured tabular datasets with nonlinear feature interactions.

### **2.4.4 Support Vector Regression with Thresholding**

To reproduce the conventional regression-based paradigm, two separate support vector regression models with radial basis function kernels were trained using the original 10-feature Kachuee feature set.<sup>7</sup> One model predicted systolic blood pressure and the second predicted diastolic blood pressure. The training pipeline included median imputation followed by standardization. Hyperparameters were selected using 5-fold subject-level GroupKFold cross-validation over regularization and kernel-width settings.

Because support vector regression has substantial memory requirements at large sample sizes, the training set was reduced to approximately one segment per subject for this baseline. Continuous systolic and diastolic predictions were then converted into ACC/AHA blood pressure categories using threshold rules. Probabilistic scores for AUROC analysis were derived from the signed distance to the systolic hypertension threshold.

### **2.4.5 One-Dimensional Convolutional Neural Network**

A one-dimensional convolutional neural network (1D-CNN) was developed for direct classification from raw ECG and PPG waveforms.<sup>6</sup> The network consisted of four sequential convolutional blocks with progressively increasing channel depth (32, 64, 128, and 128 filters). Each block contained two Conv1D layers (kernel size 7 in the first three blocks and 5 in the final block) with batch normalization, ReLU activation, and 2× max-pooling. The convolutional feature extractor was followed by adaptive average pooling and a two-layer multilayer perceptron classification head with 0.3 dropout. The final network contained approximately 397,000 trainable parameters.

Training was performed for 15 epochs using the AdamW optimizer (learning rate  $1 \times 10^{-3}$ , weight decay  $1 \times 10^{-4}$ ) with a cosine-annealing learning-rate schedule and a batch size of 256. Weighted cross-entropy loss with class weights [0.900, 2.136, 0.704] was used to counter the hypertensive over-representation, and the model checkpoint with the highest validation macro F1 score was retained for final testing. Training was performed on a single NVIDIA RTX 4070 GPU.

## 2.5 Evaluation

Primary evaluation focused on classification performance on the common held-out test set consisting of 2,398 subjects and 25,687 waveform segments, which was identical across all five models. Reported metrics included overall accuracy, defined as the proportion of correctly classified segments across all three classes, and macro F1 score, calculated as the unweighted mean of the class-specific F1 scores to account for class imbalance. Standard classification measures including precision, recall, and confusion matrix analysis were also examined to characterize class-specific performance and common misclassification patterns.

Because the principal clinical objective was hypertension detection, additional binary screening metrics were reported for the hypertensive class. Hypertensive AUROC was computed by collapsing the three-class output into a one-vs-rest formulation using the predicted probability of the hypertensive class against the corresponding ground-truth labels and integrating the resulting receiver operating characteristic curve. Hypertensive recall quantified the proportion of true hypertensive segments correctly identified, while the false-negative rate ( $1 - \text{recall}$ ) represented the clinically relevant miss rate for screening applications.

For the support vector regression baseline, regression-side error metrics were also reported to enable direct comparison with prior literature. These included mean absolute error (MAE) for systolic blood pressure, diastolic blood pressure, and mean arterial pressure. In addition to predictive performance, model interpretability was assessed using feature importance rankings for tree-based models and gradient-based saliency maps for the CNN to identify waveform regions most influential to classification decisions.

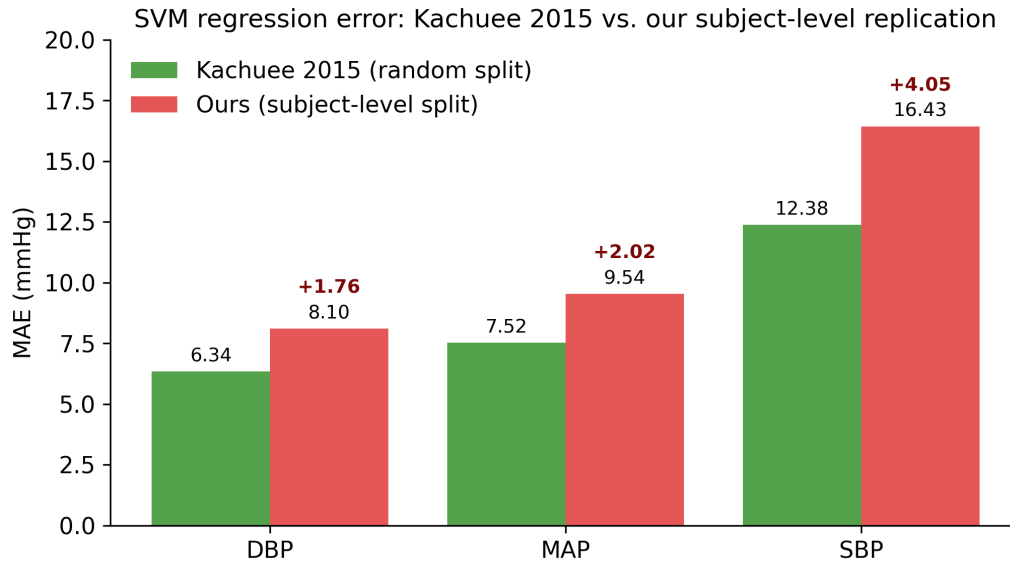
## 3. Results

### 3.1 Reproduction of the Kachuee SVM regression baseline

To contextualize the proposed classification framework, we first reproduced the Support Vector Regression (SVR) approach of Kachuee et al. using subject-level data partitioning. Regression performance was evaluated using mean absolute error (MAE) for systolic BP, diastolic BP, and mean arterial pressure (MAP) and then compared with the values in the original study.

| Quantity       | Kachuee 2015<br>(random split, 4,254 records) | Ours<br>(subject-level split, 11,987 records) |
|----------------|-----------------------------------------------|-----------------------------------------------|
| DBP MAE (mmHg) | 6.34                                          | 8.10                                          |
| MAP MAE (mmHg) | 7.52                                          | 9.54                                          |
| SBP MAE (mmHg) | 12.38                                         | 16.43                                         |

**Table 1. SVM regression MAE: original paper vs. our subject-level replication.**



**Figure 2.** Comparison of support vector regression mean absolute error between Kachuee et al. (2015) and the present subject-level replication. Errors were higher under subject-level splitting for DBP, MAP, and SBP, consistent with the increased difficulty of cross-subject generalization.

Our subject-level replication produced higher MAE across all three quantities, with absolute increases of 1.76 mmHg for DBP, 2.02 mmHg for MAP, and 4.05 mmHg for SBP relative to the original report.

### 3.2 Three-class hypertension classification

All five models were evaluated on the common held-out test set consisting of 25,687 waveform segments from 2,398 subjects. Performance was assessed using overall accuracy, macro F1 score, hypertensive AUROC, hypertensive recall, and false-negative rate.

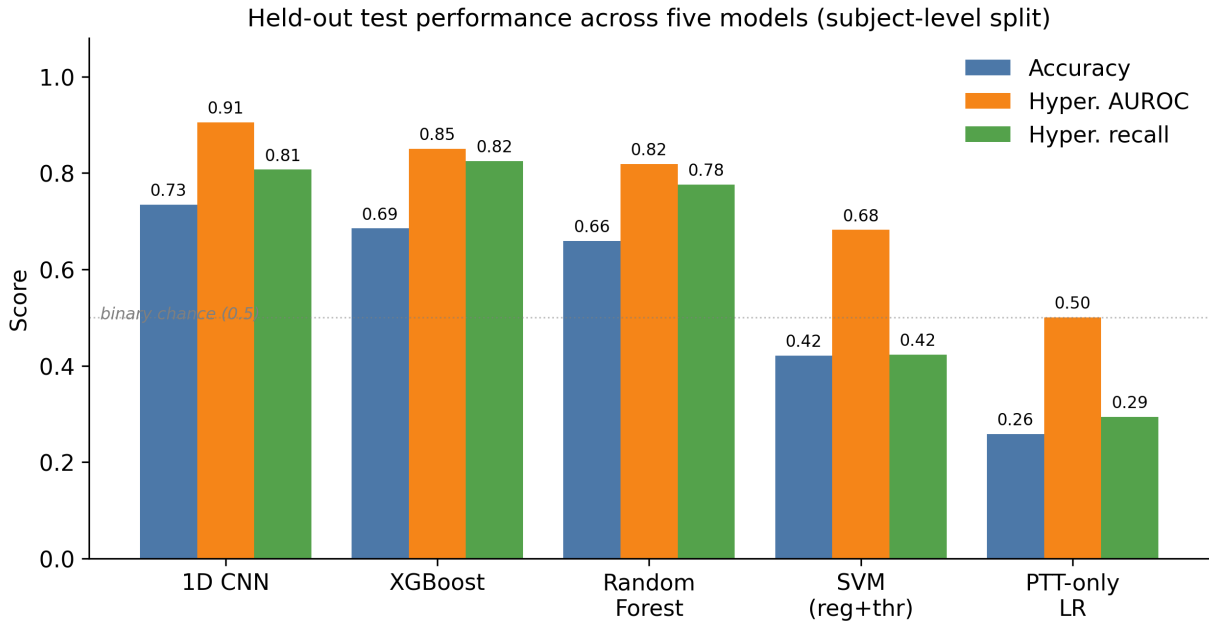
| Model                          | Accuracy | Macro-F1 | Hyper. AUROC | Hyper. recall | FNR   |
|--------------------------------|----------|----------|--------------|---------------|-------|
| 1D CNN (raw PPG+ECG)           | 0.734    | 0.686    | 0.905        | 0.807         | 0.193 |
| XGBoost (20 features)          | 0.685    | 0.583    | 0.850        | 0.825         | 0.175 |
| Random Forest (20 features)    | 0.659    | 0.583    | 0.819        | 0.776         | 0.224 |
| SVM regression + AHA threshold | 0.421    | 0.414    | 0.682        | 0.423         | 0.577 |
| PTT-only logistic regression   | 0.258    | 0.236    | 0.500        | 0.294         | 0.706 |

**Table 2. Held-out test performance, 25,687 segments from 2,398 subjects.**

The 1D CNN trained directly on raw (PPG, ECG) segments achieves a hypertensive AUROC of 0.905, compared to 0.682 for the Kachuee-style SVM regressor whose continuous SBP/DBP outputs are post-hoc thresholded into AHA classes. Accuracy improves by 31 percentage points (0.421  $\rightarrow$  0.734) and hypertensive recall by 38 points (0.423  $\rightarrow$  0.807). On the same data and the same held-out test subjects, framing the problem as classification rather than regression converts a barely-above-chance hypertension detector into a strong one.

XGBoost trained on 20 morphological and timing features (PPG pulse widths and slopes, ECG QRS duration, HR, RR variability, PTT triplet, AI, LASI, S1–S4 areas) reaches AUROC 0.850 and the highest hypertensive recall in the table (0.825), narrowly beating the CNN on recall while losing on AUROC and accuracy. This shows that most of the CNN's advantage is recoverable with handcrafted features when those features go beyond the original Kachuee ten and when the problem is framed as classification.

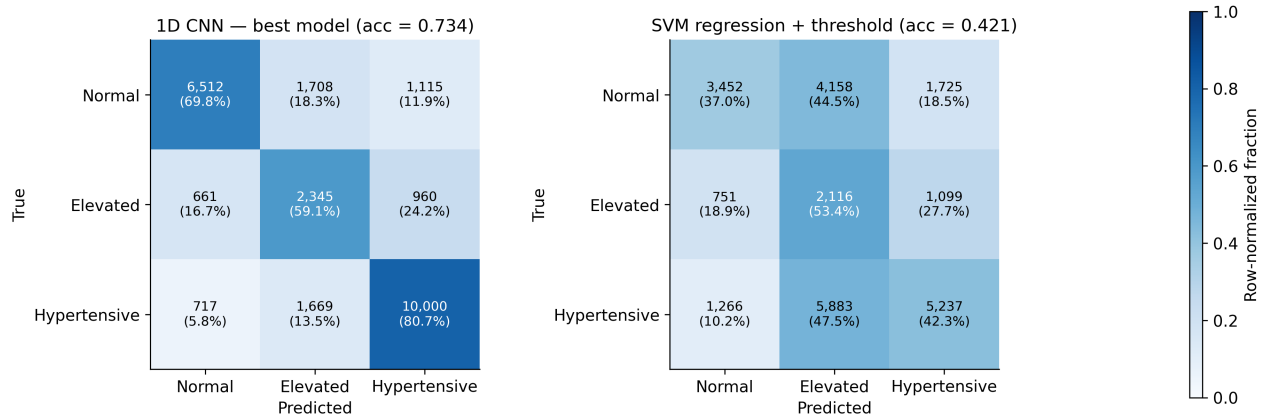
A logistic regression on PTT alone yields AUROC 0.500 — exactly chance for hypertensive detection. This is consistent with the well-known calibration limitation of PTT-based BP estimation: PTT carries strong intra-subject but weak inter-subject BP information, so once train and test subjects are disjoint, PTT loses its discriminative power. Kachuee's calibration-free claim relies on the additional PPG-morphology features; our experiments confirm that those morphological features are doing essentially all of the cross-subject work.



**Figure 3.** Held-out test performance across five models. The 1D-CNN achieved strongest overall performance, while XGBoost produced comparable results and the highest hypertensive recall.

The 1D CNN achieved the strongest overall performance, with an accuracy of 0.734, macro F1 of 0.686, and hypertensive AUROC of 0.905. XGBoost was the strongest feature-based model and achieved the highest hypertensive recall (0.825). The regression-plus-threshold SVM baseline and PTT-only logistic regression performed substantially worse across all metrics.

Confusion matrices on held-out test (n = 25,687 segments, 2,398 subjects)



**Figure 4.** Row-normalized confusion matrices on the held-out test set for the best-performing 1D CNN and the support vector regression plus threshold baseline. The CNN shows strong diagonal performance with highest recall for hypertensive segments, whereas the regression-based baseline exhibits substantial confusion across classes, particularly misclassification of hypertensive cases as elevated.

The confusion matrices demonstrate stronger diagonal performance for the 1D CNN, particularly for hypertensive segments, whereas the regression-based baseline showed substantial cross-class confusion.

### 3.3 Best-model error analysis

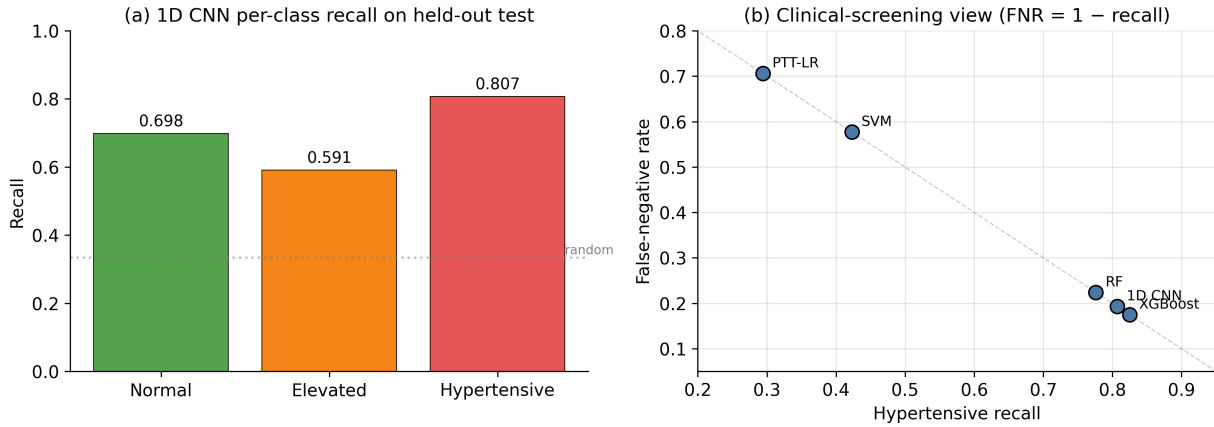
To further characterize the best-performing model, per-class recall was examined for the 1D CNN on the held-out test set.

| Class        | Recall |
|--------------|--------|
| Normal       | 0.698  |
| Elevated     | 0.591  |
| Hypertensive | 0.807  |

**Table 3.** Per-class recall of the 1D CNN on the held-out test set; hypertensive performance is markedly higher than the other two classes.

The CNN achieved recall values of 0.698 for Normal, 0.591 for Elevated, and 0.807 for Hypertensive segments. The corresponding false-negative rate for hypertensive detection was 0.193.





**Figure 5.** Classification performance for hypertension detection on the held-out test set. (a) Per-class recall of the 1D CNN, showing highest sensitivity for hypertensive segments and lowest recall for the intermediate elevated class. (b) Clinical screening comparison of all models using hypertensive recall versus false-negative rate, where lower false-negative rates and higher recall indicate better screening performance.

Among all classes, Elevated showed the lowest recall, while hypertensive recall remained high relative to competing models.

### 3.4 Feature attribution and CNN saliency

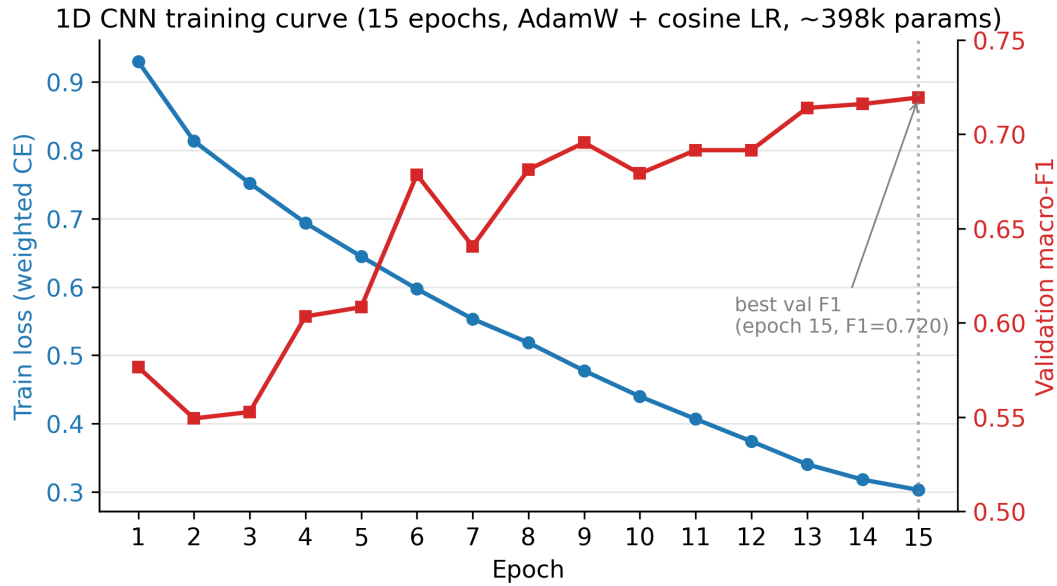
Feature attribution analysis was performed for the tree-based models, and gradient-based saliency was used to interpret the CNN.

Permutation importance for XGBoost identified the most influential predictors as PPG pulse width at 50% amplitude, ECG QRS duration, heart rate, PPG rise time, large-artery stiffness index, PPG decay time, PPG amplitude ratio, and RR interval variability. Random Forest produced a similar ranking with minor differences.

For the CNN, gradient-based saliency was distributed broadly across the waveform window in both ECG and PPG channels rather than concentrated at a single landmark, indicating that classification decisions were informed by multi-cycle waveform morphology.

### 3.5 Training Convergence

Training behavior of the 1D-CNN was monitored across 15 epochs using weighted training loss and validation macro F1 score.



**Figure 6.** Training performance of the 1D CNN over 15 epochs. Training loss decreased steadily under cosine-annealed learning rate, while validation macro-F1 improved monotonically into the final epochs.

Training loss decreased steadily, while validation macro F1 improved and reached a maximum of 0.720 at epoch 15. The best-performing checkpoint was retained for test evaluation.

## 4. Discussion

### 4.1 Principal Finding

The principal finding of this study is that framing cuff-less blood pressure assessment as a direct three-class classification problem substantially outperforms the conventional regression-then-threshold paradigm. Across all reported classification metrics, direct classifiers exceeded the performance of the replicated Kachuee-style support vector regression baseline (Table 2; Figure 3). On the same held-out subjects, hypertensive AUROC increased from 0.682 for the regression pipeline to 0.905 for the 1D CNN and 0.850 for XGBoost (Table 2). Overall accuracy improved from 0.421 to 0.734, while hypertensive recall increased from 0.423 to 0.807. These findings indicate that the choice of learning objective — classification versus regression — had a greater effect on final performance than the specific model architecture alone.

### 4.2 Why Regression-then-Threshold Underperforms

Two factors likely explain the weaker performance of the regression-based approach. First, regression models optimize a continuous error objective across the full blood pressure range, whereas the downstream clinical task depends on discrete decision thresholds. Small numerical errors near category boundaries may change the predicted class, while equivalent errors far from those thresholds have little practical consequence. Direct classification models can therefore allocate model capacity toward the regions of greatest clinical relevance.

Second, the observed regression error was large relative to the width of the intermediate Elevated category. In our replication, systolic MAE remained 16.43 mmHg under subject-level evaluation (Table 1; Figure 2), while the Elevated category spans only a narrow systolic interval.

When prediction error approaches or exceeds class width, post-hoc thresholding becomes inherently unstable. This likely contributed to the substantial class confusion observed in the regression baseline, particularly the frequent misclassification of hypertensive segments as elevated (Figure 4).

### **4.3 Implications of the PTT-Only Result**

The PTT-only logistic regression model performed at chance level for hypertensive detection (AUROC = 0.500), suggesting that pulse transit time alone provides limited cross-subject discriminatory information in this dataset (Table 2; Figure 3). This result is consistent with the known dependence of the PTT–blood pressure relationship on subject-specific factors such as vascular stiffness, age, body size, and baseline autonomic tone. While pulse transit time may be useful for tracking relative blood pressure changes within an individual, its absolute mapping appears insufficient for population-level classification when training and testing subjects are fully separated.

Feature attribution analysis further supports this interpretation. In the XGBoost model, the strongest predictors were waveform morphology and timing variables such as pulse width, QRS duration, heart rate, and rise or decay characteristics, whereas none of the PTT variants ranked among the top contributors (Section 3.4). Collectively, these findings suggest that future calibration-free systems may benefit more from richer multimodal waveform characterization than from reliance on pulse transit time alone.

### **4.4 Comparison with Prior Work**

The regression-side replication demonstrated that our implementation was consistent with the original Kachuee framework, preserving the same qualitative error ordering of DBP < MAP < SBP (Table 1). However, all regression errors increased under the stricter subject-level protocol (Figure 2). This difference highlights the importance of evaluation design in cuff-less blood pressure research. Segment-level random splits may overestimate real-world performance when multiple segments from the same subject appear in both training and test sets, allowing models to exploit subject-specific baselines that would not be available in deployment.

Beyond replication, the present study extends prior work by showing that strong hypertension detection performance remains achievable under a more rigorous evaluation framework when the task is reformulated as direct classification. The superior diagonal structure of the CNN confusion matrix and stronger hypertensive recall further support this conclusion (Figure 4; Figure 5).

### **4.5 Limitations**

Several limitations should be considered.

First, the MIMIC-II source population consists primarily of intensive care unit patients and may not reflect ambulatory or community screening populations. Accordingly, the reported AUROC should be interpreted as evidence of learnable signal rather than direct deployment performance in wearable settings (Table 2).

Second, labels were derived from the median ABP value within individual 8-second segments. Clinical hypertension is typically defined using repeated measurements over longer time horizons, and segment-level labels may therefore reflect transient hemodynamic variation rather

than sustained disease status.

Third, the dataset does not include demographic or anthropometric covariates such as age, sex, height, or weight, all of which may influence the physiological relationship between waveform features and blood pressure. Inclusion of such variables could improve cross-subject generalization.

Fourth, several landmark-based features were affected by limited visibility of the diastolic notch at the 125 Hz sampling frequency. Median imputation preserved sample size, but alternative approaches such as improved landmark detection or higher-resolution data may further strengthen feature-based baselines.

Fifth, our subject-level partitioning was enforced at the level of UCI (part, record) tuples rather than original MIMIC-II patient identifiers, which the public release does not preserve. If a single MIMIC-II patient contributed multiple records to the UCI dataset, those records could in principle land in different splits. We therefore cannot guarantee strict patient-level disjointedness; however, the protocol still prevents the more common form of leakage from repeated segments of the same record.

Finally, the strongest direct classifiers were trained using either the expanded 20-feature set or raw waveforms, whereas the regression baseline used the original 10-feature representation. Although this reflected the intended comparison between classical and expanded formulations, additional matched-feature experiments would further isolate the independent effects of task framing and feature design.

## **4.6 Ethical Considerations**

Several ethical considerations apply to the development and potential deployment of cuff-less hypertension detection models. With respect to data privacy, all signals were obtained from the publicly released UCI Cuff-Less Blood Pressure Estimation dataset, which is derived from the de-identified MIMIC-II waveform database. No protected health information or direct patient identifiers were used at any stage of this study; all analyses were conducted under the data-use terms of the public release.

With respect to model bias and generalizability, the source population is restricted to intensive care unit patients and lacks demographic covariates such as age, sex, race, and body habitus. Models trained on this distribution may therefore not generalize equitably to ambulatory, pediatric, geriatric, or under-represented populations. Reporting performance only at the aggregate level can mask subgroup-specific failure modes; future work that includes demographically annotated cohorts should report performance stratified by these covariates before any deployment is considered.

With respect to clinical implications, the best classifier in this study still missed 19.3% of hypertensive segments on the held-out test set. A false-negative rate of this magnitude is incompatible with unsupervised diagnostic use; the appropriate framing is screening or triage to flag individuals for confirmatory cuff-based measurement. Wearable deployment would also raise additional concerns around device calibration drift over time, alarm fatigue from frequent segment-level predictions, and the potential for over-medicalization if continuous monitoring outputs are surfaced to patients without appropriate clinical context. We therefore consider these results evidence of feasibility rather than a direct path to clinical decision-making, and any

translational use should be developed in partnership with regulatory and clinical-validation pathways.

#### **4.7 Implications**

Three practical recommendations emerge from this work. First, cuff-less blood pressure models should be evaluated using subject-level splits whenever repeated measurements per subject are present. Second, when the intended downstream decision is categorical, such as hypertension screening, direct classification may be more appropriate than numerical regression followed by thresholding. Third, clinically relevant metrics such as hypertensive recall and false-negative rate should be reported alongside AUROC, as discrimination alone does not fully capture screening utility (Table 2; Figure 5).

#### **4.8 Future Directions**

Future work should include external validation on non-ICU wearable datasets to assess generalizability beyond the MIMIC-derived population. Additional studies should also examine calibration performance, operating-point selection, and matched-feature comparisons that isolate the effects of feature engineering versus task formulation. Finally, integrating demographic covariates or longitudinal subject information may further improve robust cuff-less hypertension detection across diverse populations.

### **5. Conclusion**

This study evaluated whether cuff-less BP assessment is better approached as a direct classification task rather than the conventional regression-then-threshold framework. Using synchronized ECG and PPG signals from the UCI Cuff-Less Blood Pressure Estimation dataset, five machine learning models were compared under strict subject-level splits to assess cross-subject generalization. The results demonstrated that direct classification methods substantially outperformed the regression-based baseline, with the 1D CNN achieving the strongest overall performance and XGBoost providing a solid feature-based alternative.

These findings suggest that clinically meaningful BP categorization may be a more practical objective than precise numerical estimation for wearable hypertension screening, as it aligns the learning objective with real screening and triage decisions. The poor performance of the PTT-only model indicates that pulse transit time alone provides limited cross-subject discriminatory power, whereas broader waveform morphology and multimodal temporal patterns contain substantially better predictive information. The 1D-CNN achieved the best discrimination by learning directly from raw signals, while XGBoost performed well by using engineered features with greater interpretability. Future work should focus on external validation in more diverse populations, improving identification of intermediate BP classes, and ultimately on deployment strategies for wearable monitoring systems that respect the ethical and clinical constraints discussed above.

### **6. Code Availability**

All code and instructions are available in the following GitHub repository:  
<https://github.com/pa-arth/bme-ml-bp-classification>

## References

1. Whelton, P. K., Carey, R. M., Aronow, W. S., Casey, D. E., Jr., Collins, K. J., Dennison Himmelfarb, C., et al. (2018). 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA guideline for the prevention, detection, evaluation, and management of high blood pressure in adults. *Hypertension*, 71(6), e13–e115. <https://doi.org/10.1161/HYP.0000000000000065>
2. Kachuee, M., Kiani, M. M., Mohammadzade, H., & Shabany, M. (2015). Cuff-less high-accuracy calibration-free blood pressure estimation using pulse transit time. In *2015 IEEE International Symposium on Circuits and Systems (ISCAS)* (pp. 1006–1009). IEEE.
3. Kachuee, M., Kiani, M. M., Mohammadzade, H., & Shabany, M. (2015). *Cuff-Less Blood Pressure Estimation [Dataset]*. UCI Machine Learning Repository. <https://doi.org/10.24432/C5B602>
4. Salman, H. A., Kalakech, A., & Steiti, A. (2024). Random forest algorithm overview. *Babylonian Journal of Machine Learning*, 2024, 1–10.
5. Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794. <https://doi.org/10.1145/2939672.2939785>
6. Kiranyaz, S., Avci, O., Abdeljaber, O., Ince, T., Gabbouj, M., & Inman, D. J. (2021). 1D convolutional neural networks and applications: A survey. *Mechanical Systems and Signal Processing*, 151, 107398. <https://doi.org/10.1016/j.ymssp.2020.107398>
7. Ahmad, F., Isa, N. A. M., Hussain, Z., Osman, M. K., & Sulaiman, S. N. (2015). Support vector machine-based optimized decision threshold adjustment strategy for classifying imbalanced data. *Knowledge-Based Systems*, 85, 40–57. <https://doi.org/10.1016/j.knosys.2015.04.012>
8. Rainio, O., Teuho, J., & Klén, R. (2024). Evaluation metrics and statistical tests for machine learning. *Scientific Reports*, 14(1), 6086. <https://doi.org/10.1038/s41598-024-56706-x>
9. Jones, D. W., Ferdinand, K. C., Taler, S. J., Johnson, H. M., Shimbo, D., Abdalla, M., et al. (2025). 2025 AHA/ACC/AANP/AAPA/ABC/ACCP/ACPM/AGS/AMA/ASPC/NMA/PCNA/SGIM guideline for the prevention, detection, evaluation and management of high blood pressure in adults: A report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*, 152(11), e114–e218. <https://doi.org/10.1161/CIR.0000000000001356>
10. Sadiq, A. A., Othman, N. B., Abdul Jamil, M. M., Youseffi, M., Denyer, M., Wan Zakaria, W. N., & Md Tomari, M. R. (2018). Fourth-order Butterworth active bandpass filter design for single-sided magnetic particle imaging scanner. *Journal of Telecommunication, Electronic and Computer Engineering*, 10(1–17), 17–21.